



FUTURE NOTES II

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When will AI and Smart Robotics Reshape the Global Economy?

Summary

Technologies historically unfold slowly. They begin with excitement and heavy investment, then drift through a long phase of integration before finally delivering broad productivity gains. Robert Solow's famous observation that *"you can see the computer age everywhere but in the productivity statistics"* reflected exactly this long lag between invention and measurable economic impact.

Economists like Carlota Pérez have documented how general-purpose technologies such as steam & rail, electricity, and early IT followed extended installation-to-deployment timelines lasting decades.

AI stands out for its speed. Modern generative AI moved from early experimentation (2018–2020) to widespread business deployment by 2022–2023, implying just a **3–4 year gap** between invention and visible economic use — roughly **an order of magnitude faster** than past general-purpose technologies. Agentic AI takes longer to implement but is likely to show meaningful macro-economic impact within **5-6 years**.

A major reason lies in **coordination dynamics**. Human teams hit natural limits: adding people boosts **communication overhead** and **coordination costs** (a core insight of **Brooks's Law**), meaning productivity often stalls or even drops past a small team size. By contrast, **AI systems lack those biological constraints**. Digital agents can be deployed at scale with minimal onboarding and overhead, and their internal coordination can be **engineered rather than negotiated**. Scaling productive capacity therefore becomes a matter of **compute and architecture**, not human coordination, transforming what was once a social bottleneck into an engineering one.

The physical AI frontier follows fast too. Embodied AI — robots and autonomous systems — still faces **material constraints** (hardware, energy, logistics). Yet, these machines inherit AI's ability to learn autonomously and share that learning collectively. Autonomous vehicles are approaching large-scale practical deployment, and general-purpose humanoids could become

operational at scale within the next few years. For embodied AI, **transition periods to large macro-economic impact of 6–12 years**, far shorter than the typical multi-decade lags for past physical technologies.

These trends imply **higher structural growth and lower inflation** than past eras, as abundant, scalable intelligence expands productive capacity. Gains that once took decades to emerge may now unfold within the span of business cycles. This can create a surprisingly attractive market for risk assets over several decades.

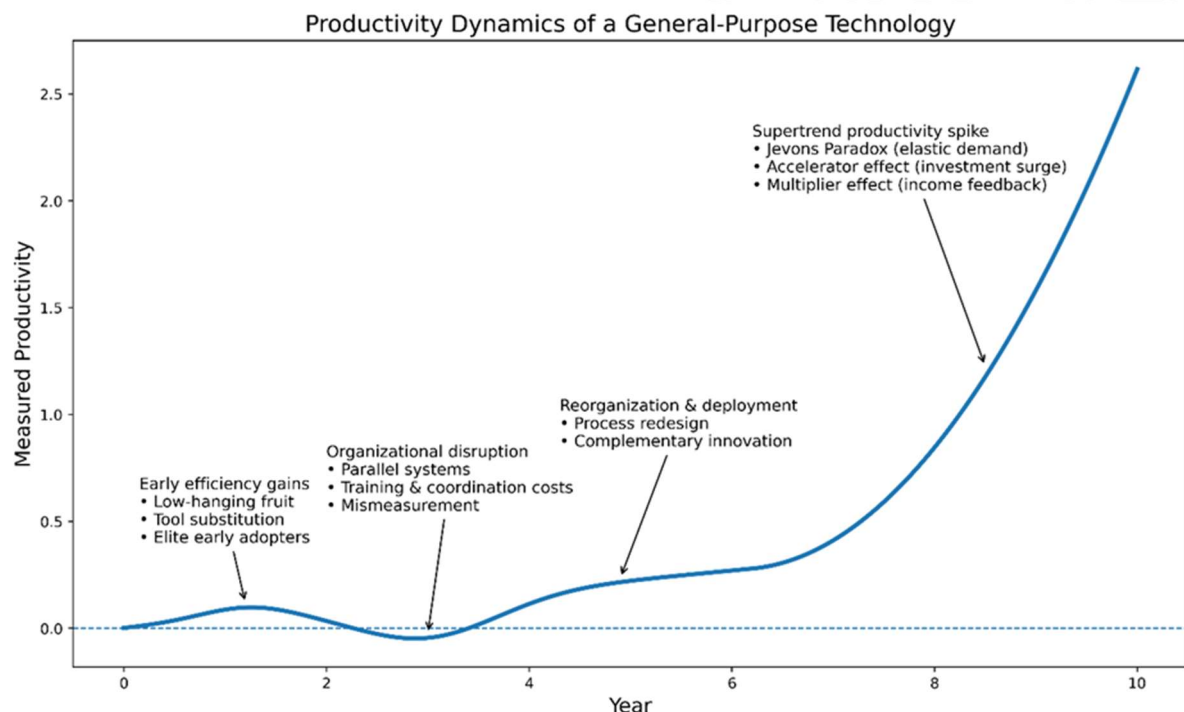
The Productivity Paradox

Imagine you're at a dinner party and someone announces they've invented a device that will change the world. Most guests may very well nod politely, then swiftly return to discussing the weather or last week's football match. Historically, this is how we greet great technologies: with a mild, fleeting curiosity rather than a profound realization of the upheaval to come.

And the indifference could last for long. For decades, economists were left scratching their heads over a famous observation by Robert Solow: "You can see the computer age everywhere but in the productivity statistics." Despite silicon chips finding their way into every toaster and filing cabinet in the land, the actual measured growth of the economy remained stubbornly indifferent for a quarter century. This puzzle became known as the **Productivity Paradox**.

The reason for this isn't that technology is a dud, but rather that "General-Purpose Technologies"—things like steam, electricity, or the internet—tend to follow a rather wobbly **J-curve**.

STYLISTIC MODEL OF TIME-TO-ECONOMIC IMPACT OF CARE GENERAL-PURPOSE TECHNOLOGIES



Source: Tvede, Lars. 2050 — When Machines Do the Work, and Humans Do the Living. Forthcoming 2026.

You start with heavy investment and frantic enthusiasm, followed by a long, awkward period where the gains are invisible to the data. Perhaps the macro-economic data even get worse before they get better.

The Smithian Struggle: Retooling the World

Why is this so? - Innovation triggers co-called **Schumpeterian Growth** (after Joseph Schumpeter), but for a technology to actually move the macro-economic needle, it must transition into what economists call **Smithian Growth** (named for Adam Smith). This is the messy phase where innovation is woven into the fabric of everyday life. To get there, factories must be retooled, supply chains redesigned, and workers retrained. This is a clumsy, frustrating, and expensive upheaval, and meanwhile, productivity can actually *sag*.

Firms often waste time trying to bolt new tools onto old processes instead of rebuilding from the ground up. Legacy systems clash with new ones, and along the way, time is spent producing nothing sellable.

But this isn't failure; it's incubation. Eventually, complementary innovations emerge, workflows simplify, and the gains move from theoretical to unmistakable.

Mapping the Pérez-Gaps

To understand the tempo of these shifts, we look to **Carlota Pérez**, a Venezuelan-British economist who noted that technological revolutions generally arrive in a two-act structure: an early, expensive **Installation** phase and a much later, profitable **Deployment** phase.

Historically, the gap between these acts has been remarkably long:

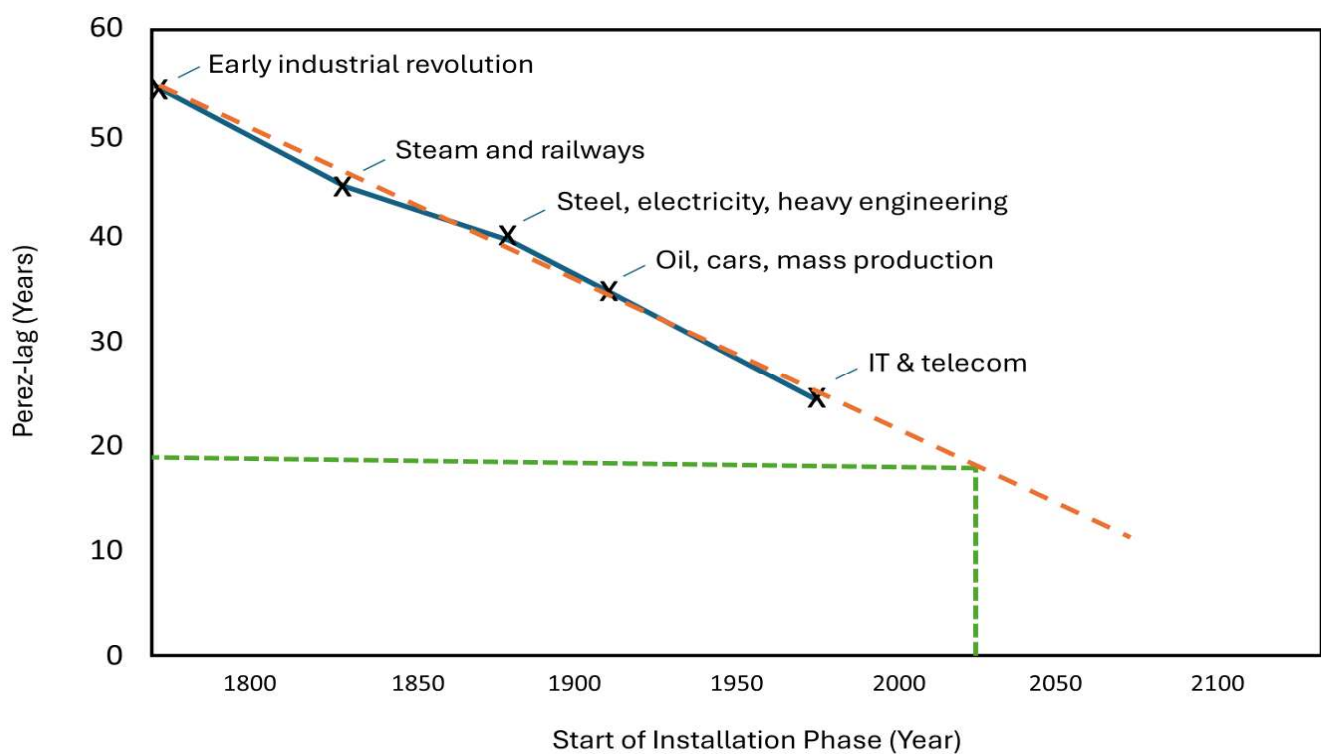
- **Industrial Revolution (Mechanization, Water Power):** Took roughly **50–60 years** from initial installation before broad economic benefits spread across the economy.
- **Steam and Railways:** Transitioned from early infrastructure build-out to widespread deployment in about **40–50 years**.
- **Steel, Electricity & Heavy Engineering:** Required approximately **35–45 years** before the new technologies reshaped industrial capacity at scale.
- **Oil, Automobiles & Mass Production:** Took around **30–40 years** to diffuse widely and deliver broad productivity gains.

- **Information & Telecommunications:** Lingered roughly **20–30 years** before the fruits of full deployment became ripe.

Interestingly, the lag **lag between installation and deployment** across successive general-purpose technology waves has been **shrinking linearly over time**.

The graph below plots the historical Perez-lags plus an extrapolation (dotted orange line). The dotted green lines show how this extrapolation predicts a Perez-lag by 2026 to be **~19 years**.

DEVELOPMENT IN THE PEREZ-GAPS OVER TIME 1775 – 1990, PLUS EXTRAPOLATION



Source: Tvede, Lars. 2050 — When Machines Do the Work, and Humans Do the Living. Forthcoming 2026.

If that historical pattern still held, the breakthroughs in large language models and AI — which crystallized around 2022 — would only begin showing up clearly in economic statistics in the early 2040s.

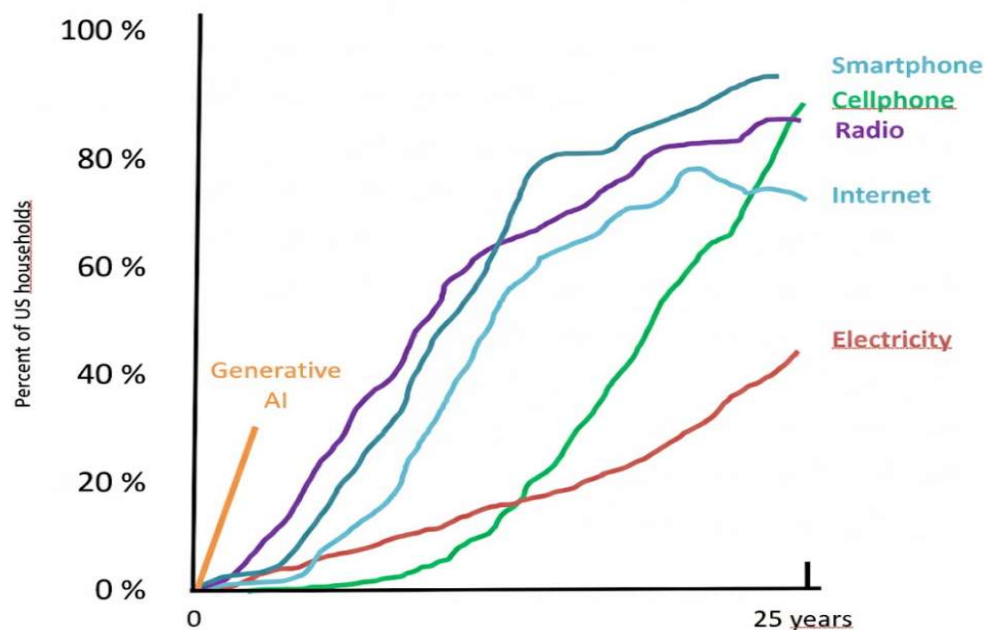
But that isn't what we see.

Disembodied Generative AI: The Short-Lag Revolution

We can already observe that there is something materially different about the Perez-lag for AI. Let's first look at "disembodied AI"—intelligence that lives in software and data centers, but not in mobile devices such as cars, robots, and drones.

In the U.S., the Installation phase for disembodied AI began in earnest between 2018 and 2020 with the rise of transformers and foundation models. By 2023, generative AI had already leapt out of the laboratory and into the daily workflows of coders, designers, and researchers. The take-off was faster than for any previous product type:

US USER TAKE-OFF SPEED FOR NEW TECHNOLOGIES



Source: Financial Times

Why so fast? First, the product didn't require any new physical devices, was amazing, and worked on the first try for everyone. Download the app — 10 seconds. Ask it a question — an amazing answer in another 10 seconds. Boom!

What we have seen very recently is that the gap between the opening act and everyday utility for generative AI was perhaps **3–4 years**—a staggering deviation from the 20 years suggested by the extrapolated Pérez-lag curve.

This is most evident in software development, where jobs have become harder to find even as total output has exploded. In other words, **coding productivity has *already* exploded:**

FOUR INDICATORS OF GROWTH IN SOFTWARE OUTPUT

The past year has seen an explosion in coding productivity

Year-on-year growth in various indicators of coding output



Sources: Domain Name Industry Brief; SensorTower; GitHub. GitHub data is quarterly pushes
FT graphic: John Burn-Murdoch / @jburnmurdoch
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Source: Financial Times

Agentic AI: From Reactive Prompts to Autonomous Action

Building on disembodied generative AI, agentic systems represent the next leap: autonomous AI agents that plan, reason, decide, and execute multi-step tasks with minimal human oversight—handling complex workflows like supply-chain optimization or R&D pipelines. Adoption is accelerating dramatically:

surveys from 2025–2026 show 35–79% of organizations with some agentic AI use or pilots, with forecasts for 40% of enterprise apps integrating task-specific agents by end-2026 and mass consumer reach in the same year. A likely Perez-lag for this is **5-6 years** as compared to use of generative AI in LLMs, it is more domain-specific and thus involves detailed analysis of local business processes etc.

Beating Brooks' Law — Why AI Isn't Constrained Like People

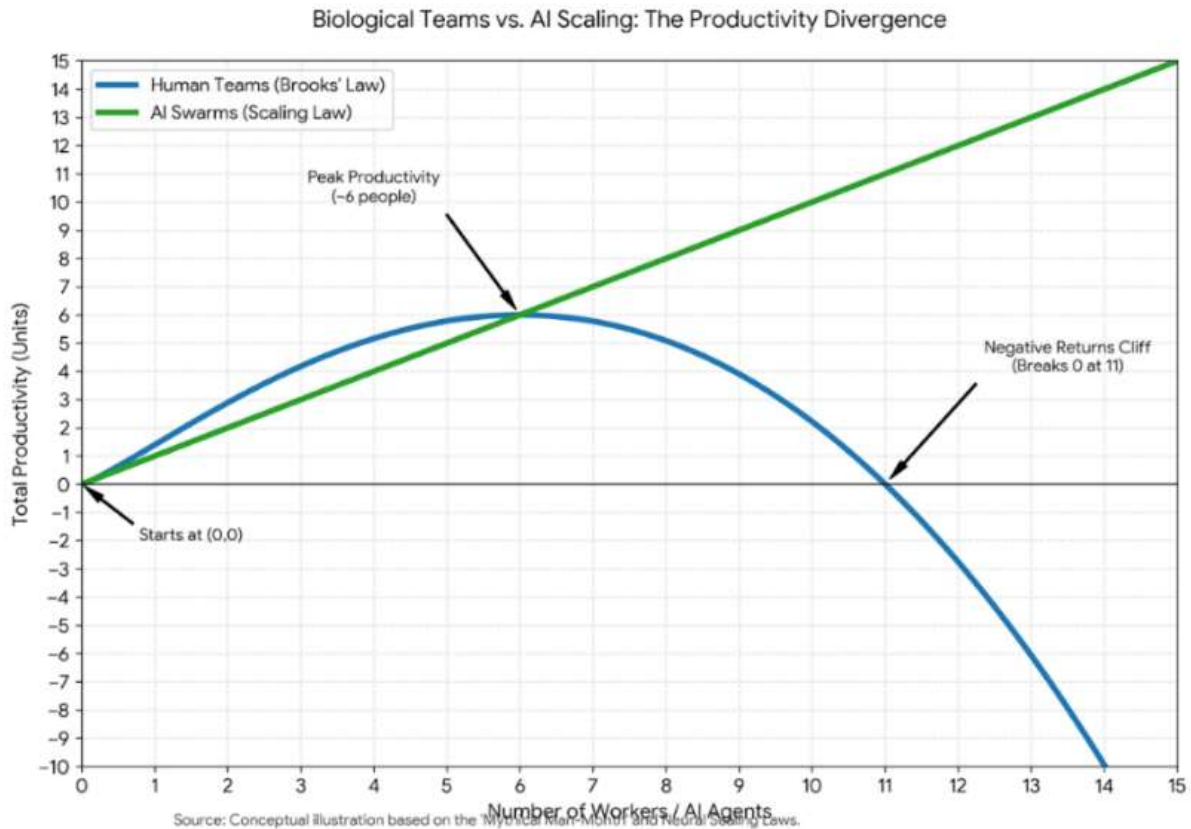
One aspect of the speedy impact of disembodied AI is often overlooked: **Brooks' Law**.

This states that *adding manpower to a late software project often makes it later*. The insight is simple but deep — every additional human introduces more *communication channels* and coordination overhead, so beyond a modest size, adding people slows progress instead of speeding it up. Empirical research bolsters this: in software and similar work, teams perform best at around **five to seven people** — enough brains to share the load but not so many that coordination bogs down. Statistics show that once a team grows beyond about **10–11 people**, productivity can actually **decline** because communication overhead rises faster than actual output.

Here is where this gets interesting, because **AI swarms can largely sidestep this constraint**. Of course, coordination also matters for AI swarms such as agentic AI — but computer protocols and architectures can be *designed* to streamline this. Instead of endless meetings going nowhere, we can build structured pipelines where each agent knows its role and how to hand off to the next — like an orchestra where every musician knows their part and no one insists on tuning their own violin mid-symphony.

Human organizations will still feel like cocktail parties — warm, unpredictable, and occasionally chaotic. Well-designed AI collectives can feel more like the Swiss train system — efficient, purposeful, and surprisingly coherent. Understanding that difference is essential for grasping where growth, productivity, and economic power will come from in the decades ahead.

BROOKS' LAW FOR HUMANS AND AI MODELS



Source: Tvede, Lars. 2050 — When Machines Do the Work, and Humans Do the Living. Forthcoming 2026.

The partial cancellation of Brooks' Law is one of the reasons that AI scales so well economically.

Embodied AI: Intelligence Meets Matter

How about AI embedded in mobile objects? Here, the **physical world still matters** — robots, autonomous vehicles, and intelligent machines involve motors, factories, mining, and energy grids.

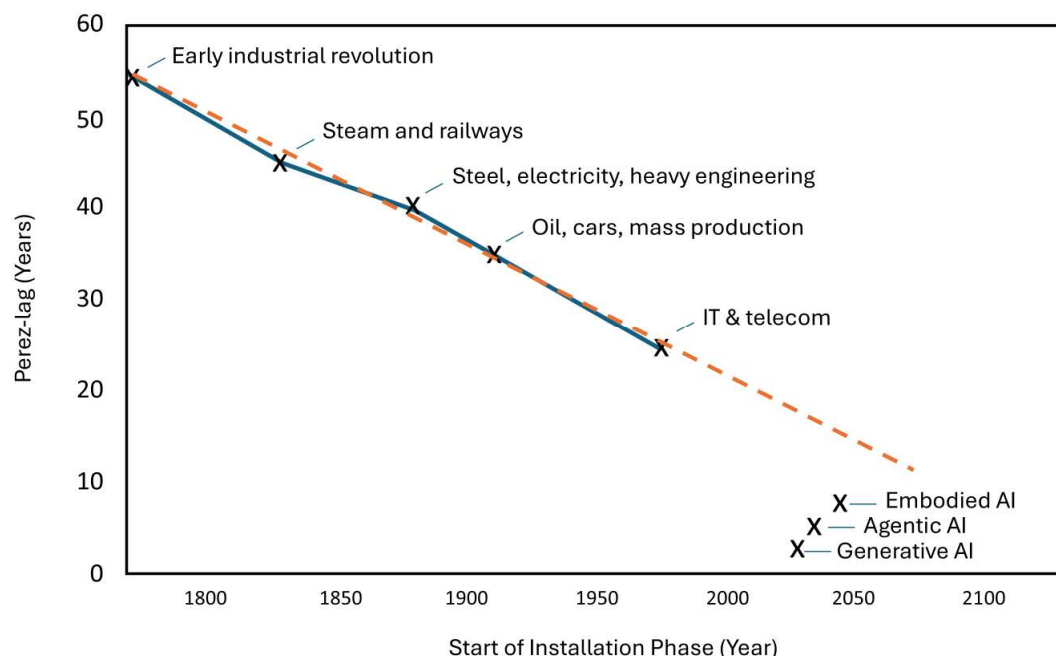
Yet embodied AI is not merely another layer of automation. These systems **inherit the exponential improvements of their software counterparts** through learning, adaptation, and continual refinement. They don't simply

execute predefined motions; they *improve* with deployment, optimizing routines and integrating into workflows with increasing sophistication. Think of them less like heavy machinery and more like **clever specialists** asking, "How can I help?"

For autonomous vehicles, practical reliability was observed in **2025**. For general-purpose humanoid robots — machines capable of meaningful work in real environments — early large-scale deployment may arrive by **2027**. Instead of the long, 15–20-year lags seen in past revolutions, embodied AI may take **6–12 years** from large-scale installation to significant economic impact — slower than disembodied AI, but *dramatically faster than historical norms for general purpose technologies*.

If current trajectories hold, we can expect embodied AI to begin **materially reshaping major sectors of the global economy throughout the 2030s**.

ESTIMATED/ PREDICTED PEREZ-LAGS VERSUS EXTRAPOLATED HISTORICAL TREND 1775 – 1990



Source: Tvede, Lars. 2050 — When Machines Do the Work, and Humans Do the Living. Forthcoming 2026.

What Does This Mean?

Standard growth models assume steady, incremental innovation lifting per-capita output only after decades. As we have argued here, AI obliterates that assumption. No prior technology has diffused this fast. The Perez-gap for Generative AI is **3-4 years**, for agentic AI perhaps **5-6 years**, and for embodied AI probably **6-12 years**.

All these are moving far faster than anything seen before, and the broad effect is likely to be **higher trend growth and lower inflation** than many forecasts assume. In short, a type of **goldilocks economy** creating a **potentially very attractive backdrop for most asset prices**.

The question isn't whether AI will materially reshape the global economy — **it is already doing so**.

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